TP2- Implement Advanced STP

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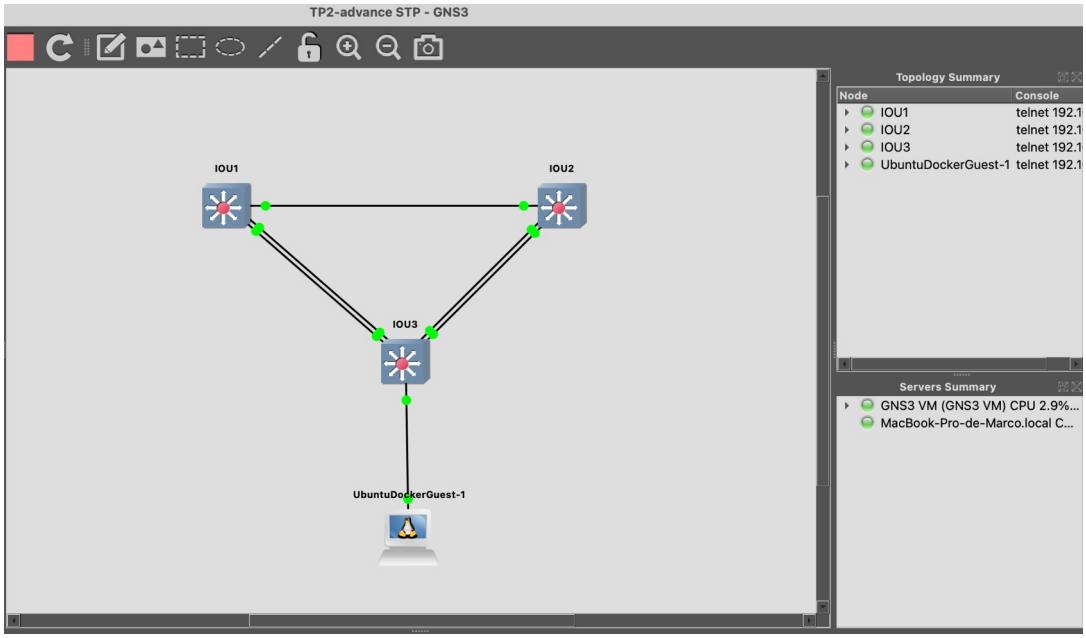
Topology



Bmarco11 — IOU1 — telnet 192.168.71.128 50

```
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
end

D1#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#interface range e1/0-3,e2/0-3, e3/0-3
D1(config-if-range)#shutdown
D1(config-if-range)#
*Oct 26 13:06:15.689: %LINK-5-CHANGED: Interface Ethernet1/0, changed state to administratively down
*Oct 26 13:06:15.708: %LINK-5-CHANGED: Interface Ethernet1/1, changed state to administratively down
*Oct 26 13:06:15.708: %LINK-5-CHANGED: Interface Ethernet1/2, changed state to administratively down
*Oct 26 13:06:15.708: %LINK-5-CHANGED: Interface Ethernet1/3, changed state to administratively down
*Oct 26 13:06:15.708: %LINK-5-CHANGED: Interface Ethernet2/0, changed state to administratively down
*Oct 26 13:06:15.708: %LINK-5-CHANGED: Interface Ethernet2/1, changed state to administratively down
D1(config-if-range)#
*Oct 26 13:06:15.708: %LINK-5-CHANGED: Interface Ethernet2/2, changed state to administratively down
*Oct 26 13:06:15.713: %LINK-5-CHANGED: Interface Ethernet2/3, changed state to administratively down
*Oct 26 13:06:15.713: %LINK-5-CHANGED: Interface Ethernet3/0, changed state to administratively down
*Oct 26 13:06:15.713: %LINK-5-CHANGED: Interface Ethernet3/1, changed state to administratively down
*Oct 26 13:06:15.722: %LINK-5-CHANGED: Interface Ethernet3/2, changed state to administratively down
*Oct 26 13:06:15.722: %LINK-5-CHANGED: Interface Ethernet3/3, changed state to administratively down
*Oct 26 13:06:16.696: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to down
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/1, changed state to down
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/2, changed state to down
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/3, changed state to down
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to down
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/1, changed state to down
D1(config-if-range)#
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/2, changed state to down
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/3, changed state to down
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/0, changed state to down
*Oct 26 13:06:16.717: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/1, changed state to down
*Oct 26 13:06:16.722: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/2, changed state to down
*Oct 26 13:06:16.722: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/3, changed state to down
D1(config-if-range)#exit
D1(config)#interface range e0/0-3
D1(config-if-range)#switchport trunk encapsulation dot1q
D1(config-if-range)#switchport mode trunk
D1(config-if-range)#no
*Oct 26 13:09:44.742: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/2, changed state to down
D1(config-if-range)#no shutdo
*Oct 26 13:09:47.763: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/2, changed state to up
D1(config-if-range)#no shutdown
D1(config-if-range)#exit
D1(config)#vlan 2
D1(config-vlan)#secondvlan
                ^
% Invalid input detected at '^' marker.

D1(config-vlan)#name secondvlan
D1(config-vlan)#exit
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 130×43

```
A1(config)#interface range e1/1-3, e2/0-3, e3/0-3
A1(config-if-range)#shutdown
A1(config-if-range)#e
*Oct 26 13:26:49.662: %LINK-5-CHANGED: Interface Ethernet1/1, changed state to administratively down
*Oct 26 13:26:49.663: %LINK-5-CHANGED: Interface Ethernet1/2, changed state to administratively down
*Oct 26 13:26:49.669: %LINK-5-CHANGED: Interface Ethernet1/3, changed state to administratively down
*Oct 26 13:26:49.669: %LINK-5-CHANGED: Interface Ethernet2/0, changed state to administratively down
*Oct 26 13:26:49.669: %LINK-5-CHANGED: Interface Ethernet2/1, changed state to administratively down
*Oct 26 13:26:49.678: %LINK-5-CHANGED: Interface Ethernet2/2, changed state to administratively down
A1(config-if-range)#exit
A1(config)#
*Oct 26 13:26:49.678: %LINK-5-CHANGED: Interface Ethernet2/3, changed state to administratively down
*Oct 26 13:26:49.678: %LINK-5-CHANGED: Interface Ethernet3/0, changed state to administratively down
*Oct 26 13:26:49.678: %LINK-5-CHANGED: Interface Ethernet3/1, changed state to administratively down
*Oct 26 13:26:49.687: %LINK-5-CHANGED: Interface Ethernet3/2, changed state to administratively down
*Oct 26 13:26:49.687: %LINK-5-CHANGED: Interface Ethernet3/3, changed state to administratively down
*Oct 26 13:26:50.664: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/1, changed state to down
*Oct 26 13:26:50.665: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/2, changed state to down
*Oct 26 13:26:50.670: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/3, changed state to down
*Oct 26 13:26:50.670: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to down
*Oct 26 13:26:50.670: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/1, changed state to down
*Oct 26 13:26:50.683: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/2, changed state to down
*Oct 26 13:26:50.684: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/3, changed state to down
A1(config)#
*Oct 26 13:26:50.684: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/0, changed state to down
*Oct 26 13:26:50.684: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/1, changed state to down
*Oct 26 13:26:50.693: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/2, changed state to down
*Oct 26 13:26:50.693: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/3, changed state to down
A1(config)#interface range e0/0-3, e1/0
A1(config-if-range)#switchport trunk encapsulation dot1q
A1(config-if-range)#
A1(config-if-range)#switchport mode trunk
A1(config-if-range)#no
*Oct 26 13:29:02.360: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to down
*Oct 26 13:29:02.360: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to down
A1(config-if-range)#no shutd
*Oct 26 13:29:05.368: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
*Oct 26 13:29:05.372: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to up
A1(config-if-range)#no shutdown
A1(config-if-range)#exit
A1(config)#vlan 2
A1(config-vlan)#name secondvlan
A1(config-vlan)#exit
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 134×56

```
IOU2#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
IOU2(config)#hostname D2
D2(config)#spanning-tree mode rapid-pvst
D2(config)#line con 0
D2(config-line)#exec-timeout 0 0
D2(config-line)#logging synchronous
D2(config-line)#exit
D2(config)#interface range e1/0-3,e2/0-3, e3/0-3
D2(config-if-range)#shutdown
D2(config-if-range)#
*Oct 26 13:18:32.039: %LINK-5-CHANGED: Interface Ethernet1/0, changed state to administratively down
*Oct 26 13:18:32.048: %LINK-5-CHANGED: Interface Ethernet1/1, changed state to administratively down
*Oct 26 13:18:32.048: %LINK-5-CHANGED: Interface Ethernet1/2, changed state to administratively down
*Oct 26 13:18:32.048: %LINK-5-CHANGED: Interface Ethernet1/3, changed state to administratively down
*Oct 26 13:18:32.048: %LINK-5-CHANGED: Interface Ethernet2/0, changed state to administratively down
*Oct 26 13:18:32.057: %LINK-5-CHANGED: Interface Ethernet2/1, changed state to administratively down
D2(config-if-range)#exit
*Oct 26 13:18:32.057: %LINK-5-CHANGED: Interface Ethernet2/2, changed state to administratively down
*Oct 26 13:18:32.057: %LINK-5-CHANGED: Interface Ethernet2/3, changed state to administratively down
*Oct 26 13:18:32.057: %LINK-5-CHANGED: Interface Ethernet3/0, changed state to administratively down
*Oct 26 13:18:32.064: %LINK-5-CHANGED: Interface Ethernet3/1, changed state to administratively down
*Oct 26 13:18:32.064: %LINK-5-CHANGED: Interface Ethernet3/2, changed state to administratively down
*Oct 26 13:18:32.064: %LINK-5-CHANGED: Interface Ethernet3/3, changed state to administratively down
*Oct 26 13:18:33.046: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to down
*Oct 26 13:18:33.056: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/1, changed state to down
*Oct 26 13:18:33.056: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/2, changed state to down
*Oct 26 13:18:33.056: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/3, changed state to down
*Oct 26 13:18:33.056: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to down
*Oct 26 13:18:33.061: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/1, changed state to down
D2(config-if-range)#exit
*Oct 26 13:18:33.061: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/2, changed state to down
*Oct 26 13:18:33.061: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/3, changed state to down
*Oct 26 13:18:33.061: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/0, changed state to down
*Oct 26 13:18:33.069: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/1, changed state to down
*Oct 26 13:18:33.069: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/2, changed state to down
*Oct 26 13:18:33.069: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/3, changed state to down
D2(config-if-range)#exit
D2(config)#interface range e0/0-3
D2(config-if-range)#switchport trunk encapsulation dot1q
D2(config-if-range)#switchport mode trunk
D2(config-if-range)#
*Oct 26 13:19:31.760: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to down
*Oct 26 13:19:31.761: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/3, changed state to down
D2(config-if-range)#
*Oct 26 13:19:34.764: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
*Oct 26 13:19:34.766: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/3, changed state to up
D2(config-if-range)#no shutdown
D2(config-if-range)#exit
D2(config)#vlan 2
D2(config-vlan)#name secondvlan
D2(config-vlan)#exit
D2(config)#interface vlan1
D2(config-if)#ip address 10.0.0.2 255.0.0.0
D2(config-if)#no shutdown
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
 state to down
A1(config-if-range)#no shutd
*Oct 26 13:29:05.368: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed
 state to up
*Oct 26 13:29:05.372: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed
 state to up
A1(config-if-range)#no shutdown
A1(config-if-range)#exit
A1(config)#vlan 2
A1(config-vlan)#name secondvlan
A1(config-vlan)#exit
A1(config)#interface vlan 1
A1(config-if)#ip address 10.0.0.3 255.0.0.0
A1(config-if)#no shutdown
A1(config-if)#exit
A1(config)#
*Oct 26 13:31:07.627: %LINK-3-UPDOWN: Interface Vlan1, changed state to up
*Oct 26 13:31:08.636: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state
 to up
A1(config)#exit
A1#sh
*Oct 26 13:37:33.261: %SYS-5-CONFIG_I: Configured from console by console
A1#show spanning-tree root

                                          Root    Hello Max Fwd
Vlan                Root ID               Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         24577 aabb.cc00.0100       100     2   20  15  Et0/1
VLAN0002         28674 aabb.cc00.0100       100     2   20  15  Et0/1
A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
  Root ID    Priority    24577
             Address     aabb.cc00.0100
             Cost        100
             Port        2 (Ethernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.0300
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Altn BLK 100       128.3    Shr
Et0/3               Altn BLK 100       128.4    Shr
Et1/0               Desg FWD 100       128.5    Shr


A1#show spanning-tree blockedports

Name                 Blocked Interfaces List
-------------------- ------------------------------------
VLAN0001             Et0/2, Et0/3
VLAN0002             Et0/2, Et0/3

Number of blocked ports (segments) in the system : 4

A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
  Root ID    Priority    24577
             Address     aabb.cc00.0100
             Cost        100
             Port        2 (Ethernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.0300
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Altn BLK 12        128.3    Shr
Et0/3               Altn BLK 100       128.4    Shr
Et1/0               Desg FWD 100       128.5    Shr


A1#show spanning-tree blockedports

Name                 Blocked Interfaces List
-------------------- ------------------------------------
VLAN0001             Et0/2, Et0/3
VLAN0002             Et0/2, Et0/3

Number of blocked ports (segments) in the system : 4

A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
             Address     aabb.cc00.0300
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Altn BLK 12        128.3    Shr
Et0/3               Altn BLK 100       128.4    Shr
Et1/0               Desg FWD 100       128.5    Shr


A1#show spanning-tree blockedports

Name                 Blocked Interfaces List
-------------------- ------------------------------------
VLAN0001             Et0/2, Et0/3
VLAN0002             Et0/2, Et0/3

Number of blocked ports (segments) in the system : 4

A1#show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                  Root ID           Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         24577 aabb.cc00.0100       100     2   20  15  Et0/1
VLAN0002         28674 aabb.cc00.0100       100     2   20  15  Et0/1
A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
Et0/2               Altn BLK 12        128.3    Shr
Et0/3               Altn BLK 100       128.4    Shr
Et1/0               Desg FWD 100       128.5    Shr


A1#show spanning-tree vlan 2

VLAN0002
  Spanning tree enabled protocol rstp
  Root ID    Priority    28674
             Address     aabb.cc00.0100
             Cost        100
             Port        2 (Ethernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32770  (priority 32768 sys-id-ext 2)
             Address     aabb.cc00.0300
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Altn BLK 12        128.3    Shr
Et0/3               Altn BLK 100       128.4    Shr
Et1/0               Altn BLK 100       128.5    Shr


A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Altn BLK 12        128.3    Shr
Et0/3               Altn BLK 100       128.4    Shr
Et1/0               Altn BLK 100       128.5    Shr


A1#debug spanning-tree events
Spanning Tree event debugging is on
A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)#interface e0/0
A1(config-if)#no shutdown
A1(config-if)#
A1(config-if)#shutdown
A1(config-if)#
*Oct 26 14:13:49.093: %LINK-5-CHANGED: Interface Ethernet0/0, changed state to administrat
ively down
*Oct 26 14:13:50.101: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed
 state to down
A1(config-if)#end
A1#
*Oct 26 14:13:59.554: %SYS-5-CONFIG_I: Configured from console by console
A1#undegub all
        ^
% Invalid input detected at '^' marker.

A1#undebug all
All possible debugging has been turned off
A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
%Warning: portfast should only be enabled on ports connected to a single
 host. Connecting hubs, concentrators, switches, bridges, etc... to this
 interface  when portfast is enabled, can cause temporary bridging loops.
 Use with CAUTION

%Portfast has been configured on Ethernet0/0 but will only
 have effect when the interface is in a non-trunking mode.
A1(config-if)#no shutdown
A1(config-if)#exit
A1(config)#
*Oct 26 14:15:39.815: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Oct 26 14:15:40.831: RSTP(1): initializing port Et0/0
*Oct 26 14:15:40.832: RSTP(1): Et0/0 is now designated
*Oct 26 14:15:40.832: RSTP(2): initializing port Et0/0
*Oct 26 14:15:40.832: RSTP(2): Et0/0 is now designated
*Oct 26 14:15:40.841: RSTP(1): transmitting a proposal on Et0/0
*Oct 26 14:15:40.841: RSTP(2): transmitting a proposal on Et0/0
A1(config)#
*Oct 26 14:15:41.321: RSTP(2): transmitting a proposal on Et0/0
*Oct 26 14:15:41.321: RSTP(1): transmitting a proposal on Et0/0
A1(config)#
*Oct 26 14:15:41.840: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed
 state to up
A1(config)#
*Oct 26 14:15:43.334: RSTP(2): transmitting a proposal on Et0/0
*Oct 26 14:15:43.334: RSTP(1): transmitting a proposal on Et0/0
A1(config)#
*Oct 26 14:15:45.344: RSTP(2): transmitting a proposal on Et0/0
*Oct 26 14:15:45.344: RSTP(1): transmitting a proposal on Et0/0
A1(config)#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
Vlan                Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
VLAN0001            Desg FWD 100       128.1    Shr
VLAN0002            Desg FWD 100       128.1    Shr
A1#show spanning-tree detail | section Ethernet0/0
          from Ethernet0/0
 Port 1 (Ethernet0/0) of VLAN0001 is designated forwarding
          from Ethernet0/0
 Port 1 (Ethernet0/0) of VLAN0002 is designated forwarding
A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)#interface e0/0
A1(config-if)#spanning-tree portfast trunk
%Warning: portfast should only be enabled on ports connected to a single
 host. Connecting hubs, concentrators, switches, bridges, etc... to this
 interface  when portfast is enabled, can cause temporary bridging loops.
 Use with CAUTION

A1(config-if)#no shutdown
A1(config-if)#exit
A1(config)#end
A1#
*Oct 26 14:22:09.754: %SYS-5-CONFIG_I: Configured from console by console
A1#show spanning-tree interface e0/0

Vlan                Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
VLAN0001            Desg FWD 100       128.1    Shr Edge
VLAN0002            Desg FWD 100       128.1    Shr Edge
A1#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 85×24

```
D2(config-if-range)#exit
D2(config)#
*Oct 26 14:04:32.092: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1,
anged state to up
*Oct 26 14:04:32.092: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/3,
anged state to up
D2(config)#end
D2#
*Oct 26 14:04:36.849: %SYS-5-CONFIG_I: Configured from console by console
D2#show span root

                                          Root    Hello Max Fwd
Vlan                  Root ID             Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         24577 aabb.cc00.0100       100     2   20  15  Et0/0
VLAN0002         28674 aabb.cc00.0100       100     2   20  15  Et0/0
D2#show spanning-tree detail | include VLAN0002
 VLAN0002 is executing the rstp compatible Spanning Tree protocol
 Port 1 (Ethernet0/0) of VLAN0002 is root forwarding
 Port 2 (Ethernet0/1) of VLAN0002 is designated forwarding
 Port 3 (Ethernet0/2) of VLAN0002 is designated forwarding
 Port 4 (Ethernet0/3) of VLAN0002 is designated forwarding
 Port 5 (Ethernet1/0) of VLAN0002 is designated forwarding
D2#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 85×24

```
                                          Root    Hello Max Fwd
Vlan                  Root ID             Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         24577 aabb.cc00.0100       100     2   20  15  Et0/0
VLAN0002         28674 aabb.cc00.0100       100     2   20  15  Et0/0
D2#show spanning-tree detail | include VLAN0002
 VLAN0002 is executing the rstp compatible Spanning Tree protocol
 Port 1 (Ethernet0/0) of VLAN0002 is root forwarding
 Port 2 (Ethernet0/1) of VLAN0002 is designated forwarding
 Port 3 (Ethernet0/2) of VLAN0002 is designated forwarding
 Port 4 (Ethernet0/3) of VLAN0002 is designated forwarding
 Port 5 (Ethernet1/0) of VLAN0002 is designated forwarding
D2#show span root | include VLAN0002
VLAN0002         28674 aabb.cc00.0100       100     2   20  15  Et0/0
D2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)#interface range e1/0, e0/2
D2(config-if-range)#spanning-tree guard root
D2(config-if-range)#
*Oct 26 14:28:17.107: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port
 Ethernet1/0.
*Oct 26 14:28:17.109: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port
 Ethernet0/2.
D2(config-if-range)#exit
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 85×42

```
 Ethernet0/2.
D2(config-if-range)#exit
D2(config)#exit
D2#
*Oct 26 14:29:04.653: %SYS-5-CONFIG_I: Configured from console by console
D2#
*Oct 26 14:31:32.279: %SPANTREE-2-ROOTGUARD_BLOCK: Root guard blocking port Ethernet1
/0 on VLAN0002.
D2#show spanning-tree vlan 2

VLAN0002
  Spanning tree enabled protocol rstp
  Root ID    Priority    16386
             Address     aabb.cc00.0300
             Cost        200
             Port        1 (Ethernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32770  (priority 32768 sys-id-ext 2)
             Address     aabb.cc00.0200
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Root FWD 100       128.1    Shr
Et0/1               Desg FWD 100       128.2    Shr
Et0/2               Desg BKN*100        64.3    Shr *ROOT_Inc
Et0/3               Desg FWD 100       128.4    Shr
Et1/0               Desg BKN*100       128.5    Shr *ROOT_Inc


D2#show spanning-tree inconsistentports

Name                 Interface                Inconsistency
-------------------- ------------------------ ------------------
VLAN0002             Ethernet0/2              Root Inconsistent
VLAN0002             Ethernet1/0              Root Inconsistent

Number of inconsistent ports (segments) in the system : 2

D2#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
Et0/0       1-2
Et0/1       1-2
Et0/2       2
Et0/3       2
Et1/0       2
A1#
A1#
A1#
A1#
A1#
A1#
A1#
A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)# interface e0/1
A1(config-if)#spanning-tree bpduguard enable
A1(config-if)#
*Oct 26 14:38:29.684: %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port Et0/1 with BPDU G
uard enabled. Disabling port.
*Oct 26 14:38:29.685: %PM-4-ERR_DISABLE: bpduguard error detected on Et0/1, putting Et0/1
in err-disable state
A1(config-if)#
*Oct 26 14:38:29.688: RSTP(1): updt roles, root port Et0/1 going down
*Oct 26 14:38:29.688: RSTP(1): Et0/3 is now root port
*Oct 26 14:38:29.691: STP[1]: Generating TC trap for port Ethernet0/3
*Oct 26 14:38:30.695: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed
 state to down
A1(config-if)#
*Oct 26 14:38:31.695: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to down
A1(config-if)#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
*Oct 26 14:39:55.696: RSTP(2): transmitting a proposal on Et0/1
 --More--
*Oct 26 14:39:57.710: RSTP(2): transmitting a proposal on Et0/1
 --More--
*Oct 26 14:39:59.722: RSTP(2): transmitting a proposal on Et0/1
*Oct 26 14:40:00.144: RSTP(2): Et0/1 fdwhile Expired
*Oct 26 14:40:00.144: STP[2]: Generating TC trap for port Ethernet0/1
Et0/0       1-2
Et0/1       1-2
Et0/2       2
Et0/3       2
Et1/0       2
A1#
A1#
A1#
A1#
A1#
A1#
A1#
A1#
A1#show spanning-tree interface e0/0 detail | i BPDU
    BPDU: sent 766, received 0
    BPDU: sent 767, received 0
A1#show spanning-tree interface e0/0 detail | i BPDU
    BPDU: sent 785, received 0
    BPDU: sent 786, received 0
A1#show spanning-tree interface e0/0 detail | i BPDU
    BPDU: sent 786, received 0
    BPDU: sent 787, received 0
A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
VLAN0002
  Spanning tree enabled protocol rstp
  Root ID    Priority    16386
             Address     aabb.cc00.0300
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    16386  (priority 16384 sys-id-ext 2)
             Address     aabb.cc00.0300
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr Edge
Et0/1               Desg FWD 100       128.2    Shr
Et0/2               Desg FWD 12        128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr
Et1/0               Desg FWD 100       128.5    Shr


A1#show spanning-tree inconsistentports

Name                 Interface                Inconsistency
-------------------- ------------------------ ------------------
VLAN0001             Ethernet0/1              Loop Inconsistent

Number of inconsistent ports (segments) in the system : 1

A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
*Oct 26 14:50:22.871: RSTP(2): transmitting a proposal on Et0/1
A1#show spanning-tree inconsistentports
*Oct 26 14:50:24.883: RSTP(2): transmitting a proposal on Et0/1
A1#show spanning-tree inconsistentports

Name                 Interface                Inconsistency
-------------------- ------------------------ ------------------

Number of inconsistent ports (segments) in the system : 0

A1#
*Oct 26 14:50:26.895: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:28.904: RSTP(2): transmitting a proposal on Et0/1
*Oct 26 14:50:29.771: RSTP(2): Et0/1 fdwhile Expired
A1#
*Oct 26 14:50:30.917: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:32.923: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:34.938: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:36.947: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:38.954: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:40.966: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:42.980: RSTP(2): transmitting a proposal on Et0/1
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 90×30

```
*Oct 26 14:50:29.771: RSTP(2): Et0/1 fdwhile Expired
A1#
*Oct 26 14:50:30.917: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:32.923: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:34.938: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:36.947: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:38.954: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:40.966: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:42.980: RSTP(2): transmitting a proposal on Et0/1
A1#
*Oct 26 14:50:44.775: RSTP(2): Et0/1 fdwhile Expired
*Oct 26 14:50:44.775: RSTP[2]: Et0/1 dispute resolved
A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)#inerface e0/1
             ^
% Invalid input detected at '^' marker.

A1(config)#interface e0/1
A1(config-if)#no spanning-tree guard loop
A1(config-if)#end
A1#
*Oct 26 14:51:43.204: %SYS-5-CONFIG_I: Configured from console by console
A1#
```